4 **(a)** State, for an oscillating system, what is meant by

(i) natural frequency of vibration,

…….....

……... [1]

(ii) resonance.

…….....

…….....

……..…….…... [2]

(b) State and explain one situation where resonance is useful.

…….

…….

…….

…….....[2]

(c) A car component of mass 0.430 kg undergoes forced oscillation.

Fig. 4.1 shows how the amplitude of the oscillation varies with frequency.

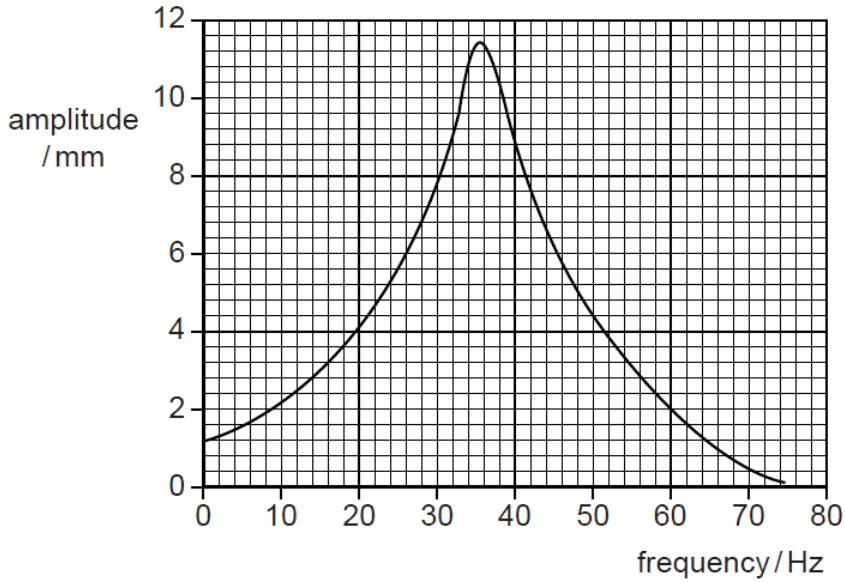

Fig. 4.1

(i) Calculate, for the component oscillating at the resonant frequency, the time interval between a maximum linear speed and the subsequent maximum linear acceleration.

time = …….. s [3]

(ii) The component is now supported on a rubber mounting.

On Fig. 4.1, draw a line to show the variation with frequency of amplitude for the supported component. [2]